

MSc Academic Portfolio — Darshitkumar Gohel

MSc Sustainable Energy Systems | FH Oberösterreich, Wels | 114 ECTS | GPA ~2.43

Purpose: Master reference for academic project highlights, technical skills, and CV bullet points drawn from MSc coursework (2022–2024). Use to populate role-specific CVs and cover letters. All metrics are simulation/model outputs from assessed coursework.

Technical Competencies

Programming & Scripting

- Python:** Pandas, NumPy, statsmodels (ARIMA/SARIMA), Matplotlib, Seaborn, Scikit-Learn
- MATLAB / Simulink:** load flow, control systems, PV-BESS simulation, time-domain dynamics
- GAMS:** LP, QP, MILP, MIQCP — power systems economics and dispatch optimization

Simulation & Design Tools

- Solar:** PVsyst V7.4.7, PV*SOL Premium 2023, AutoCAD PVcase
- Wind:** windPRO 4.1.254, WAsP 11
- Resource Data:** Meteonorm, SolarGIS, Global Solar Atlas, ERA5, MERRA-2

Methods

Domain	Methods
Meteorological	MCP, Weibull fitting, wind shear profiling, Perez/Ineichen transposition, clearness index KT
Power Systems	Dynamic Economic Dispatch, DC-OPF, N-1 contingency, MILP/MIQCP Unit Commitment, merit-order
Grid Integration	Q(U) and cosφ(P) voltage droop control, hosting capacity analysis, Newton-Raphson load flow
BESS	Cycle & calendar degradation, temporal arbitrage, SoC tracking, peak-shaving, self-consumption
Solar	Fixed-tilt and tracking simulation, bifacial gain modeling, shading loss matrices
Control	PID design, Ziegler-Nichols tuning, state-space transient analysis

Standards & Norms

IEC 61400-1 Class IA · IEC 61215 · IEC 61730 · IEC 61724 · VDE-AR-N 4105 · VDE-AR-N 4110 · EN 50160 · ISO 9613-2 · TA Lärm · EN 303-5 · NEC Article 690.12

Academic Projects

1 · Climatological Resource Assessment & Time-Series Forecasting

Courses: Data Analysis (ILV, 2.5 ECTS) · Energy Meteorology (ILV, 2.5 ECTS)

Conducted two complementary climatological resource studies — Berlin vs. Honolulu and Linz vs. New Delhi — using 43 years of historical meteorological data (1980–2022). Cleaned and preprocessed datasets in Python (Pandas/NumPy) and MATLAB, then developed ARIMA and SARIMA models to forecast 10-year GHI and wind speed trends. Delivered ranked location recommendations with quantified resource differentials and temperature-induced derating.

CV Bullets

- Forecasted solar irradiance and wind speed over a 10-year horizon by developing ARIMA and SARIMA time-series models in Python on 43-year historical datasets, projecting mean irradiance of 5.8 kWh/m²/day (Honolulu) vs. 3.0 kWh/m²/day (Berlin) to support site selection.
- Quantified a 75% higher raw solar yield in New Delhi vs. Linz in MATLAB, offsetting findings with 8–12% temperature-induced PV cell efficiency losses to deliver a calibrated comparative resource assessment.

2 · Utility-Scale Solar PV Engineering — Biratnagar, Nepal

Courses: Large-Scale Solar Power Plants (ILV, 3.0 ECTS) · Solar Power Plant Design & Engineering (UE, 1.5 ECTS)

Engineered layout design and yield optimization for a 22.5–27 MWp utility-scale solar PV plant in Biratnagar, Nepal. Developed AutoCAD PVcase layouts and ran multi-variant simulations in PVsyst V7.4.7 using Meteonorm and SolarGIS weather data. Evaluated central vs. string inverter topologies (Sungrow SG350-HX vs. SMA Sunny Central 4400 UP) and compared fixed-tilt, single-axis tracking, and bifacial configurations across three design variants. Optimal variant: bifacial + SAT achieved PR 91.46% and LCOE €0.02/kWh.

CV Bullets

- Designed a 27 MWp solar PV plant layout in AutoCAD PVcase and simulated three design variants in PVsyst; the optimal bifacial tracking configuration achieved a PR of 91.46% and LCOE of €0.02/kWh.
- Delivered a 22.5 MWp pre-feasibility study using PVsyst, SolarGIS, and Meteonorm, achieving a simulated PR of 89.6% and projected LCOE of €0.024/kWh.
- Developed a PV energy loss model quantifying thermal, cabling, and mismatch losses (18% total) and audited fire mitigation measures for NEC Article 690.12 rapid shutdown compliance (<30 V in 30 s).

3 · Solar Energy Laboratory

Course: Solar Energy — VO+UE (4.0 ECTS)

Six-module laboratory series covering solar geometry modeling, PV module characterization, off-grid and grid-tied system simulation, solar thermal collector design, and CSP plant control.

CV Bullets

- Modeled dynamic solar geometry and hourly sun-path tracking in Simulink, achieving angular error < 0.15° and GHI profile prediction within 4.5% of Meteonorm reference data.
- Performed I-V characterization and bypass diode behavior tests on PV modules, calculating a Voc temperature coefficient of -0.38%/°C and quantifying partial shading losses at 33%.
- Verified physical PV module measurements (Voc 38.3 V, Isc 9.0 A) against manufacturer datasheets via laboratory test bench.
- Sized and simulated an off-grid PV system with 24 kWh Li-ion storage in MATLAB, achieving a 98.2% annual solar fraction and limiting loss-of-load probability to under 0.5%.
- Simulated grid-tied PV-BESS energy routing and control in MATLAB/Simulink, raising local self-consumption from 35.2% to 78.4% and cutting peak grid import demand by 52.5% (4.2 kW).
- Optimized solar collector tilt in Wels, Austria (34° fixed), using a Simulink transposition model to project a 14.8% increase in annual incident energy.
- Derived flat-plate collector efficiency via quadratic regression in MATLAB: 78.2% peak efficiency, 62.4% annual solar fraction for a domestic hot water system.
- Modeled a solar thermal collector loop in Simulink with matched-flow pump control, reducing auxiliary energy consumption by 14.2% and preventing stagnation.
- Simulated a parabolic trough CSP plant and Rankine power block in MATLAB/Simulink; PID controller stabilized HTF outlet at 390°C with gross power block efficiency of 37.8%.
- Benchmarked a 15 MWth biomass CHP plant and a 1,200 m² solar thermal array during site inspection, documenting process efficiencies and EN 303-5 standards compliance.

4 · Renewables Laboratory — Commercial & Residential PV Design

Course: Renewables Laboratory (LB, 3.0 ECTS)

CV Bullets

- Designed a commercial rooftop PV system in PV*SOL, optimizing inverter-to-DC loading ratio at 1.10 to restrict annual clipping losses below 0.25% while achieving a simulated PR of 88.5%.
- Evaluated 5 residential PV-BESS design variants in PV*SOL Premium; optimal bifacial configuration reduced LCOE by 10.8% to €0.0834/kWh; non-viable options (0% IRR) explicitly flagged.
- Simulated a grid-connected residential PV system in PV*SOL, forecasting annual yield of 9,883 kWh and system PR of 88.6%.

5 · Large-Scale Wind Farm Design — Sorga (24 MW)

Course: Large-Scale Wind Power Plants (ILV, 3.0 ECTS)

Directed full pre-development engineering for the Sorga Wind Farm (5 WTGs, 118 m hub height, 24 MW). Performed MCP long-term wind speed corrections using ERA5 and MERRA-2 reanalysis data, reducing energy

projection uncertainty from 4.62% to 2.95% standard deviation. Fitted Weibull distributions ($A = 6.15$, $k = 1.298$), profiled vertical wind shear ($\alpha = 0.437$), and modeled two layout variants in windPRO. Delivered full environmental compliance package: acoustic noise (ISO 9613-2 German Interim Method), shadow flicker analysis, and IEC 61400-1 Class IA structural validation.

CV Bullets

- Engineered a 5-turbine wind farm layout in windPRO and WASP; MCP corrections using ERA5/MERRA-2 reanalysis data improved energy projection accuracy, reducing standard deviation from 4.6% to 2.9%.
 - Processed 54,000+ wind speed data points across 10 height levels (40–188 m) in Excel, fitting Weibull scale (A) and shape (k) parameters per sector to calibrate windPRO simulations.
 - Calculated sectoral wind shear exponents (α 0.11–0.62) across 12 directional bins, providing localized vertical wind profile corrections for turbine hub-height extrapolation.
 - Simulated Variant 1 (24 MW) wake losses using the N.O. Jensen Park 2 model in windPRO with site-specific air density correction (1.190 kg/m^3) to forecast AEP and optimize turbine spacing.
 - Simulated Variant 2 (21 MW, Vestas V136) wake interactions in windPRO using the N.O. Jensen model, benchmarking energy yield efficiency per installed MW against Variant 1.
 - Modeled acoustic emissions for the 5-turbine layout using ISO 9613-2:2024 German Interim Method in windPRO, verifying TA Lärm compliance at all residential receptors ($<40 \text{ dB(A)}$ at night).
 - Simulated worst-case shadow flicker impacts in windPRO, identifying a 105 hr/year threshold violation at receptor NSA1 and designing an operational curtailment schedule to restore compliance.
 - Validated IEC 61400-1 Class IA structural compliance for all 5 WTGs in windPRO, assessing peak ground acceleration (0.5 m/s^2), lightning rates, and wind shear compatibility.
-

6 · Power Systems Optimization

Course: Operations Research for Energy Systems (ILV+PT, 5.5 ECTS)

Developed a suite of mathematical optimization models in GAMS covering the full power systems economics stack: thermal dispatch, environmental dispatch, optimal power flow, unit commitment, BESS integration, and transport optimization.

CV Bullets

- Simulated a 24-hour Dynamic Economic Dispatch for a 1,400 MW system in GAMS, minimizing total fuel costs to serve 21.4 GWh daily demand and identifying the 1,143 MW evening peak as primary bottleneck.
- Integrated a 300 MWh / 60 MW BESS into a GAMS dynamic economic dispatch model, achieving system cost savings of \$80,603/day (0.77%) and reducing peak thermal generation requirements by 60 MW.
- Solved a multi-scenario DC Optimal Power Flow on a 5-bus grid in GAMS under N-1 contingency conditions; identified Line 4-5 (240 MW) as the critical bottleneck and demonstrated that wind

integration topology drives up to a 190% system cost difference.

- Formulated a Mixed-Integer Quadratic Constrained Program (MIQCP) in GAMS for Unit Commitment, proving that a 50% reduction in ramp rates increases scheduling costs by 1.51% and forces auxiliary unit shutdowns.
 - Formulated and solved an Environmental Dispatch model in GAMS, establishing a Pareto front showing a 44.6% CO₂ reduction achievable at a 35.4% system fuel cost premium.
 - Simulated multi-scenario thermal economic dispatches in GAMS; must-run constraints increase costs by 1.53%, while 100 MW of low-cost capacity addition reduces fuel costs by 15.1%.
 - Co-optimized a PV-BESS household system with flexible e-bike demand in MATLAB using linear programming, reducing daily electricity costs by 92% (€5.92 → €0.47) and verifying the equal-incremental-cost theorem at $\lambda^* = 7.92$.
 - Modeled and solved a multi-node supply chain transport problem in GAMS using LP, mapping optimal flow routes and identifying capacity-constrained bottlenecks.
-

7 · Grid Integration & Distribution Networks

Courses: Grid Integration of Renewable Power Plants (LB+VO, 4.0 ECTS) · Modern Energy Transport & Distribution (UE+VO, 5.0 ECTS)

CV Bullets

- Sized and modeled a 5 MW / 10 MWh BESS with dynamic inverter reactive power controls in MATLAB, limiting MV feeder voltage deviations by 42% and maintaining frequency excursions within VDE-AR-N 4110 grid code limits.
 - Increased distribution grid PV hosting capacity by 45% by modeling Q(U) voltage control and distributed PV systems in MATLAB, mitigating feeder overloading in compliance with VDE-AR-N 4105.
 - Reduced line losses by 12.4% via Newton-Raphson load flow simulation and voltage profile analysis in MATLAB, maintaining all bus voltages within EN 50160 limits (0.95–1.05 pu).
 - Analyzed India's energy transition pathways, outlining strategies to integrate 500 GW of non-fossil generation by 2030 and detailing grid infrastructure requirements for the 22 GW Green Energy Corridor.
-

8 · Interdisciplinary Project — PV-BESS Household Energy Management

Course: Interdisciplinary Project (PT, 3.0 ECTS)

CV Bullet

- Optimized household energy management and raised self-sufficiency to 82.5% by developing a PV-BESS simulation with cycle-aging degradation model in Python, reducing annual battery capacity fade by 36% through smart charging profiles.
-

9 · Control Engineering for Energy Systems

Course: Control Engineering for Energy Systems (LB, 3.0 ECTS)

CV Bullet

- Designed and tuned closed-loop PID controllers in MATLAB/Simulink for dynamic electrical systems, achieving settling time < 150 ms and overshoot < 4.2%.
-

10 · Financing

Course: Financing (ILV, 2.5 ECTS)

CV Bullet

- Constructed a 20-year DCF model in Excel for PV/wind asset evaluation, calculating NPV (€1.2M), IRR (6.8%), LCOE, and DSCR across varying debt-to-equity scenarios.
-

11 · Certifications

- MathWorks **MATLAB Onramp** — completed
 - MathWorks **Simulink Onramp** — completed
-

German Lebenslauf Entries (DACH-Ready)

Projektingenieur & Analyst — Windenergie & Solarsysteme

FH Oberösterreich | Wels, Österreich | 09/2023 – Heute

- **Sorga Windparkplanung (24 MW):** Windressourcenanalyse und Ertragsberechnung für einen 5-WTG-Windpark in windPRO/WAsP; MCP-Korrekturen (ERA5/MERRA-2) reduzierten Prognoseunsicherheit von 4,6 % auf 2,9 %; Weibull-Parameteranpassung ($A = 6,15$, $k = 1,298$) und Windscherungsanalyse über 10 Höhenstufen (40–188 m) auf Basis von 54.000+ Messpunkten.
- **Sorga Schall- & Schattenwurf:** Immissionsprognosen nach ISO 9613-2 (TA Lärm) mit Verifikation aller Wohnrezeptoren < 40 dB(A) nachts; Schattenwurfsimulation mit Identifikation einer Grenzwertüberschreitung (105 h/Jahr, NSA1) und Entwicklung eines Abschaltregimes zur Genehmigungsfähigkeit.
- **Sorga Strukturelle Compliance (IEC 61400-1):** Validierung von Bodenbeschleunigung ($0,5 \text{ m/s}^2$), Blitzeinschlaghäufigkeit und Windscherungskompatibilität für 5 WTGs in windPRO.
- **Nepal Utility-Scale PV (27 MWp):** Layoutplanung in AutoCAD PVcase, Ertragssimulation in PVsyst über 3 Designvarianten; optimale Bifazial-Tracker-Konfiguration erzielte PR 91,5 % und LCOE €0,02/kWh.

Analyst für Energiesystemoptimierung & Netzintegration

FH Oberösterreich | Wels, Österreich | 09/2023 – Heute

- **GAMS-Netzoptimierung:** DC-OPF-Modelle unter N-1-Ausfallbedingungen; Netzengpass Leitung 4-5 (240 MW) identifiziert; Nachweis eines 190 %igen Systemkostenunterschieds in Abhängigkeit der Windeinspeisetopologie.

- **Unit Commitment (MIQCP):** Kraftwerkseinsatzplanung mit Rampenbeschränkungen in GAMS; 300-MWh/60-MW-BESS-Integration für Arbitrage (\$80.603/Tag Ersparnis, 60-MW-Spitzenreduktion).
 - **Netzintegration (MATLAB):** Q(U)-Spannungsregelung erhöhte PV-Hosting-Kapazität um 45 %; Newton-Raphson-Lastfluss reduzierte Netzverluste um 12,4 % (EN 50160-konform).
 - **PV-BESS-Eigenverbrauchsoptimierung:** Eigenverbrauch von 35 % auf 78 % gesteigert, Peak-Shaving um 52,5 % (4,2 kW) mittels MATLAB/Simulink-Zustandsmaschinenregelung.
-

Internal Reference

Gap Analysis

Real-world gaps not directly evidenced in coursework: laboratory battery cycler data (physical aging tests); high-voltage substation design; grid-forming inverter controls; AC transient stability simulation (DigSILENT PowerFactory / PSCAD).

Mitigation paths:

1. GAMS optimization portfolio is a direct bridge to electricity market bidding and ancillary services algorithms.
2. PVcase + windPRO = commercially deployed tools — frame as industry-ready, not academic-only.
3. Adding PyBaMM electrochemical degradation modeling would materially strengthen PhD battery storage applications.

PhD / Research Angle

- **Research novelty:** Co-optimization of PVcase spatial layout with bifacial-tracking gains in monsoon/high-dust microclimates (Biratnagar); dynamic BESS + thermochemical unit commitment scheduling under high-penetration wind/solar with reanalysis uncertainty quantification.
- **Target positions:** Fraunhofer ISE (Freiburg) — Smart Grid Engineering Group; AIT Vienna — Center for Energy, EV Technologies & Power Systems.

ATS Keywords

English: Renewable Energy Systems · WindPRO · PVsyst · WAsP · GAMS · Optimal Power Flow · Dynamic Economic Dispatch · Unit Commitment · BESS Optimization · Battery Degradation Modeling · Grid Integration · Hosting Capacity Analysis · VDE-AR-N 4105 · VDE-AR-N 4110 · ISO 9613-2 · TA Lärm · MATLAB · Simulink · Python · AutoCAD PVcase · MCP · Weibull Distribution · Time-Series Forecasting · ARIMA · SARIMA

German: Energiesysteme · Netzintegration · Windparkplanung · Ertragsberechnung · Ertragsprognose · PVsyst · windPRO · GAMS-Optimierung · Lastflussberechnung · Batteriespeicher · Degradationsmodellierung · Eigenverbrauchsoptimierung · VDE-AR-N 4110 · TA Lärm · Regelungstechnik · Schattenwurfsimulation · Immissionsschutz · Schallimmission · MCP-Korrektur · Weibull-Parameteranpassung